

Estimating Statistics & Choice of Evaluation Plan -- DBMS

Definition: Cost-Based Query Optimization estimates the physical execution cost (Disk I/O, CPU cycles, Network latency) of candidate evaluation plans using System Catalog Statistics to select the minimal cost plan.

1. Detailed Technical Explanation

1. Catalog Statistics Kept by DBMS:

* n_r : Number of tuples in relation r .

2. Cost Estimation Formulas:

Selection Cost Estimation:

Join Cost Estimation (rowwise):

3. Choice of Evaluation Plan:

The query optimizer generates multiple physical evaluation plans, computes total cost = {Disk I/O Cost} + CPU Cost, and selects the plan with the minimum total cost.

2. Core Concepts & Memory Keywords

* Catalog Statistics: Metadata (n_r , b_r , $V(A,r)$) used for selectivity calculation.

3. Must-Write Points for Exams

* Disk I/O (number of block transfers) is the primary cost metric in database query evaluation.

4. Quick Recall Flow

System Catalog Stats -> Compute Selectivity -> Estimate Disk Block Accesses -> Select Plan with Lowest Total Cost